

PROJECT SUBMISSION DOSSIER

Official Submission Form and Project Documentation

1. Executive Project Overview

Project Title	Vision Beyond Sight
Team Name	Vision For All
Team Leader Name	Jiya Deshmukh
Team Members	Sandhya Vishwakama
GitHub Repository Link	https://github.com/Jiya0937/Vision-for-All.git
Demo Video Link	https://drive.google.com/file/d/1LkvBZ2Tb5SwS7NgUx7xbtavSMb8rzYUd/view?usp=drivesdk
Model Type Used	YOLOv8 + OpenCV (Hybrid/YOLO)
Dataset Source	Custom Braille Dataset + Roboflow Annotation
Dataset Link	C:\Users\jiyad\OneDrive\Documents\Desktop\Vision For All Project\backend\dataset\braille
Model Weights Link	[Insert Model Weights Link/Location Here]
Training Code Location	training/
Inference Code Location	flask_api.py

2. Detailed Project Description

BrailleAI – Vision For All is an advanced, AI-powered Braille detection and translation system developed during the hackathon using YOLOv8, OpenCV, Flask, and Node.js. The primary objective of this project is to assist visually impaired individuals by bridging the gap between physical tactile Braille patterns and the digital ecosystem. By leverage computer vision and machine learning, the system seamlessly converts physical Braille inputs into accurate, highly readable digital text.

Key Paradigm:

The system was meticulously trained on a custom Braille dataset consisting of annotated character images, and successfully integrated into a cohesive, production-ready full-stack application pipeline.

Core System Architecture Capabilities:

- **Braille Image Upload & Processing:** Robust handling of static image uploads via an interactive client interface.
- **Custom Trained YOLOv8 Inference:** High-speed, high-accuracy object detection tuned precisely for unique Braille dot configurations.
- **Real-Time Frontend/Backend Integration:** Streamlined full-stack pipeline bridging node-based operations with Python AI logic.
- **Flask-Based AI Inference API:** High-performance microservice dedicated entirely to executing model predictions.
- **Webcam Preview & Capture Functionality:** Live stream support allowing users to point their cameras directly at Braille text for instantaneous processing.
- **Interactive Premium User Interface:** A polished, responsive, and highly accessible user interface built around end-user accessibility metrics.

3. Comprehensive Technology Stack Used

Layer / Category	Technologies & Applied Scope
Frontend Components	HTML5, CSS3, JavaScript (ES6+) Designed for complete responsiveness and accessible layout parameters.
Backend Components	Node.js, Express.js, Flask (Python) Dual-runtime environment to handle real-time asset hosting and concurrent machine learning inference tasks.
AI/ML Core Frameworks	YOLOv8, OpenCV, PyTorch, Ultralytics Ecosystem Leveraged for rapid object bounding, neural network compilation, and matrix-based image manipulation.
Development Tools	VS Code, GitHub, Roboflow Annotation Suites Utilized for image annotations, dataset versioning, continuous source control, and modular coding workflows.

4. Deployment & Execution Instructions

Please execute the following operations in sequence to spin up the local development cluster:

1. Clone Repository

```
| git clone https://github.com/Jiya0937/Vision-for-All.git
```

2. Open Project Folder

```
| cd Vision-for-All
```

3. Install Python Dependencies

```
| pip install -r requirements.txt
```

4. Start Flask AI Inference API

```
| python flask_api.py
```

5. Initialize Gateway Backend Server

```
| cd backend  
| node server.js
```

6. Access Client Web Interface

```
| Open your preferred browser and navigate to: http://localhost:5000
```

5. Secondary Parameters & Asset Manifest

Sample Input Link	https://github.com/Jiya0937/Vision-for-All/tree/main/sample_input
Sample Output Link	https://github.com/Jiya0937/Vision-for-All/tree/main/sample_output
AI Tools Used	Ultralytics YOLOv8 Framework, Roboflow Computer Vision Hub
Private Model Access Instructions	No private models used. The system runs natively using open-source weights and locally run scripts.
Contact Number for Live Verification	6232150937

☒ **Mandatory Confirmation Checkbox:**

I hereby confirm that all details provided in this dossier are accurate, the local installation workflow has been verified, and the submitted source code is ready for live assessment.